

1 Static Yao-Kivelson Model

1.1 Kitaev Model and Yao-Kivelson Model

Kitaev found a way to map the strongly-interacting spin model

$$H = - \sum_{\substack{\text{x-bonds} \\ \langle jk \rangle}} J_{jk}^x \sigma_j^x \sigma_k^x - \sum_{\substack{\text{y-bonds} \\ \langle jk \rangle}} J_{jk}^y \sigma_j^y \sigma_k^y - \sum_{\substack{\text{z-bonds} \\ \langle jk \rangle}} J_{jk}^z \sigma_j^z \sigma_k^z, \quad (1.1)$$

now known as the Kitaev Model, to a free Majorana-Fermion hopping model, and therefore managed to solve the model exactly [1]. Classically, 'the ' Kitaev model is defined on a honeycomb lattice, where each coupling type $\alpha = x, y, z$ corresponds to one of the three bond orientations. However, his method directly generalizes to any tricoordinate lattice geometry [2], provided that each site has exactly one bond of each type $\alpha = x, y, z$.

One alternative to the honeycomb lattice is the star/triangle-honeycomb lattice. Instead of elementary plaquettes of size 6, it features two different types of plaquettes of size 3 and 12, respectively. When defined on this geometry, model 1.1 is known as the Yao-Kivelson Model [3]. Due to the odd length of the triangular plaquettes, this model features a groundstate that spontaneously breaks Time-Reversal Symmetry, and therefore describes a Chiral Spin Liquid.

1.2 Kitaev's Spin-Majorana Mapping

Kitaev introduced a representation of a spin- $\frac{1}{2}$ -degree of freedom (called 'spin' in the following) in terms of 4 fermionic Majorana modes. Consider the 4-dimensional Fockspace $\mathcal{H}$ of two abstract fermions a_1, a_2 satisfying the canonical anti-commutation relations

$$\left\{ a_k, a_l^\dagger \right\} = \delta_{kl} \quad \text{and} \quad \left\{ a_k, a_l \right\} = 0 \quad \text{for} \quad k, l = 1, 2. \quad (1.2)$$

They give rise to the 4 operators

$$c_{2k-1} := c_k + c_k^\dagger \quad \text{and} \quad c_{2k} := \frac{1}{i} \left(a_k - a_k^\dagger \right), \quad \text{for} \quad k = 1, 2, \quad (1.3)$$

which are commonly called Majorana modes. They satisfy

$$c_i^\dagger = c_i \quad \text{and} \quad \{c_i, c_j\} = 2\delta_{ij} \quad \text{for} \quad i, j = 1, \dots, 4, \quad (1.4)$$

which qualifies them to be called "real fermions", in contrast to the usual complex ones.

These will be used to represent the spin algebra (see sec. 2.1), i.e. construct a set $\{\sigma^\alpha\}_{\alpha=x,y,z}$ of 2-dim. hermitian operators obeying

$$\sigma^\alpha \sigma^\beta = \delta^{\alpha\beta} + i\epsilon^{\alpha\beta\gamma} \sigma^\gamma, \quad \alpha, \beta, \gamma = x, y, z. \quad (1.5)$$

For that, rename the 4 Majorana modes from $\{c_i\}_{i=1,\dots,4}$ to $\{b^x, b^y, b^z, d\}$ and define

$$\tilde{\sigma}^\alpha := ib^\alpha d, \quad \alpha = x, y, z. \quad (1.6)$$

Direct calculation shows that this definition suffices eq. 1.5 if and only if $\tilde{\sigma}^x \tilde{\sigma}^y \tilde{\sigma}^z \stackrel{!}{=} i$ holds, which is equivalent to

$$1 \stackrel{!}{=} b^x b^y b^z d =: D, \quad (1.7)$$

where we have defined the hermitian operator D . Due to $D^2 = 1$, $\mathcal{H}$ splits into the eigenspaces $\mathcal{H} = \mathcal{H}_1 \oplus \mathcal{H}_{-1}$, where the subscript denotes the respective eigenvalues of D . $\mathcal{H}_1$ is the space where eq. 1.7 is fulfilled, and is therefore called the 'physical' subspace, while $\mathcal{H}$ is called 'extended' Hilbert space. D cannot be the identity on $\mathcal{H}$, as it anti-commutes with each Majorana operator:

$$\{D, c_i\} = 0 \quad \text{for} \quad c_i = b^x, b^y, b^z, d. \quad (1.8)$$

This anti-commutation implies that each Majorana operator c_i maps 1-to-1 between $\mathcal{H}_1$ and $\mathcal{H}_{-1}$, which shows that $\dim(\mathcal{H}_1) = \dim(\mathcal{H}_{-1}) = 2$. Equation 1.8 further implies that D commutes with operators consisting of an even number of Majorana operators. In particular,

$$[D, \tilde{\sigma}^\alpha] = 0 \quad \text{for} \quad \alpha = x, y, z, \quad (1.9)$$

which means the $\tilde{\sigma}^\alpha$ preserve the physical subspace and their restriction to $\mathcal{H}_1$,

$$\sigma^\alpha := \tilde{\sigma}^\alpha|_{\mathcal{H}_1}, \quad (1.10)$$

is well-defined and finally yields the desired representation of the spin algebra.

Any Hamiltonian describing a spin can be seen as Pauli matrices acting on $\mathcal{H}_1$. It can be beneficial to replace the σ^α by $\tilde{\sigma}^\alpha$ acting on the extended Hilbert space, i.e. allowing states to have unphysical degrees of freedom. Such a state can always be projected into $\mathcal{H}_1$ using the projector

$$P = \frac{1 + D}{2} \quad (1.11)$$

to obtain a physical state. If we manage to find an H -eigenstate $|\psi\rangle$ in the extended Hilbert space $\mathcal{H}$, the projection $P|\psi\rangle \in \mathcal{H}_1$ will either be a non-normalized physical eigenstate of the same energy or the zero-vector. This follows from the fact that D and therefore also P commute with H according to eq. 1.9. Hence, in case the problem admits a solution in the extended Hilbert space, it is also solved in the physical space.

To obtain more intuition about the unphysical degrees of freedom, Kitaev used the concept of gauge transformations. Together with the identity, D is a unitary representation of the gauge group $\mathcal{G} = Z_2$ acting on the extended Hilbert space $\mathcal{H}$. This allows the characterization of the physical states $\mathcal{H}_1$ as

$$|\psi\rangle \text{ is physical} \iff |\psi\rangle \text{ is invariant under } \mathcal{G}. \quad (1.12)$$

The projector from eq. 1.11 can be interpreted as the group-average, which is in general known to project into the invariant subspace. If two states are related to each other by the application of D , they are called gauge-equivalent and they correspond to the same physical state. The converse is in general not true: Not all the states corresponding to the same physical state need to be gauge equivalent.

The mapping discussed in this section generalizes straight-forwardly to many-spin systems, where the operators now act on the product space. The gauge symmetry then becomes a local gauge symmetry at each individual site, implying that each gauge-sector contains 2^N gauge-equivalent states in the 4^N -dim. extended Hilbert space of N spins.

1.3 Application of the Mapping to the Model

Applying the spin-Majorana mapping from eq. 1.6 to a general Kitaev Hamiltonian 1.1 yields

$$H = i \sum_{\substack{\text{bonds} \\ \langle jk \rangle}} J_{jk} u_{jk}^{\alpha(j,k)} (d_j d_k) \quad \text{with} \quad u_{j,k}^\alpha := i b_j^\alpha b_k^\alpha = (u_{j,k}^\alpha)^\dagger, \quad (1.13)$$

which is grouped into d -Majoranas living on the lattice sites, and pairs of b^α -Majoranas living on the bonds. The value $\alpha(j, k) = x, y, z$ should be determined by the bond $\langle jk \rangle$, but this explicit dependence will be omitted from here on. One has to pick some fixed orientation for each bond, as each bond $\langle jk \rangle$ should uniquely identify the operators $(d_j d_k)$ and (u_{jk}^α) , even though they are each anti-symmetric under exchanging $j \leftrightarrow k$.

We now focus on lattices where each site has at most one bond of each type $\alpha = x, y, z$, and denote this requirement by $(*)$. In this case, the $u_{j,k}^\alpha$ commute among each other, which can be seen by picking two arbitrary different bonds:

Case 1: The bonds share no site. Then the Majorana modes constituting the two u -operators are 4 independent anti-commuting Majorana modes, i.e. the u -operators are independent commuting Majorana bilinears.

Else: The bonds share exactly one site, w.l.o.g. labeled by $u_{j,k}^\alpha$ and $u_{k,l}^\beta$ with $j \neq l$. Then $u_{j,k}^\alpha u_{k,l}^\beta = i^2 b_j^\alpha b_k^\alpha b_k^\beta b_l^\beta$ consists of 4 independent Majorana modes if and only if $\alpha \neq \beta$, which is guaranteed by $(*)$. The claim follows analogously to case 1.

From this, we can conclude that each $u_{j,k}^\alpha$ also commutes with the Hamiltonian 1.13: Apart from terms $u_{j,k}^\alpha$, H only consists of terms $d_j d_k$, which are Majorana bilinears independent of all the $u_{j,k}^\alpha$. With $(u_{j,k}^\alpha)^2 = 1$, this means H decomposes into symmetry-sectors labeled by the eigenvalues ± 1 of all the $u_{j,k}^\alpha$. In each symmetry sector, the u -operators can then be replaced by their eigenvalue and it remains to solve a quadratic hopping Hamiltonian, which in general possess an accessible exact solution.

However, for the gauge transformation $D_l = b_l^x b_l^y b_l^z d_l$ from eq. 1.7, one finds

$$D_l^\dagger u_{jk}^\alpha D_l = u_{jk}^\alpha \cdot \begin{cases} -1, & j = l \text{ or } j = k \\ +1, & \text{else} \end{cases}, \quad (1.14)$$

i.e. for all sites l , each $\{u_{jk}^\alpha\}$ -eigenstate is gauge equivalent to an eigenstate from a different sector, namely the sector obtained by flipping the u_{jk}^α -labels on all bonds

connected to the site l . Therefore, fixing the values of the u_{jk}^α is a practical tool to solve for the eigenstates, but it does not provide a gauge-invariant characterization of the H -eigenstates. To construct such observables, consider a sequence of lattice bonds $\gamma = (\langle j_1 k_1 \rangle, \langle j_2 k_2 \rangle, \dots, \langle j_n k_n \rangle)$ defining an operator

$$W_\gamma := \prod_{\langle jk \rangle \in \gamma} u_{jk}^{\alpha(j,k)}, \quad (1.15)$$

where $\alpha(j, k)$ is picked depending on the bond $\langle jk \rangle$ identically as in the Hamiltonian 1.13. As all the u_{jk}^α commute and square to 1, we assume w.l.o.g. no duplicate bonds in γ . According to eq. 1.14, W_γ is gauge invariant under the gauge transformation D_j on site j if and only if γ has an even number of bonds containing the site j . Geometrically, this is equivalent to γ consisting only of closed loops of bonds.

2 Additional Notes

2.1 Representations of the Pauli Algebra

If we introduce 3 hermitian operators $\{\tilde{\sigma}^x, \tilde{\sigma}^y, \tilde{\sigma}^z\}$ on a 2-dim. Hilbert space, what are the precise conditions for these to represent a spin unitarily equivalently to the usual Pauli matrices? Indeed, it is enough to check that the relation

$$\tilde{\sigma}^\alpha \tilde{\sigma}^\beta = \delta^{\alpha\beta} + i\epsilon^{\alpha\beta\gamma} \tilde{\sigma}^\gamma, \quad \alpha, \beta, \gamma = x, y, z. \quad (2.1)$$

holds according to the following lemma.

Lemma 1. *All 2-dim. hermitian representations of the Pauli algebra are unitarily equivalent, i.e. if some hermitian 2-dim. operators $\{\tilde{\sigma}^x, \tilde{\sigma}^y, \tilde{\sigma}^z\}$ suffice equation 2.1, there exists a unitary transformation U such that*

$$\tilde{\sigma}^\alpha = U^\dagger \sigma^\alpha U \quad \text{for } \alpha = x, y, z, \quad (2.2)$$

where σ^α are the usual Pauli matrices in some fixed basis.

Hence, if a set of matrices $\{\tilde{\sigma}^\alpha\}$ suffices these requirements, we can transform any spin Hamiltonian H to $\tilde{H}$ by replacing all the σ^α by $\tilde{\sigma}^\alpha$. This just amounts to picking a certain basis in the abstract Hilbert space.

Proof. We construct a basis in which the matrix representation of the $\tilde{\sigma}^\alpha$ agrees with the usual Pauli matrices. From eq. 2.1 for $\alpha = \beta$ it follows that all the $\tilde{\sigma}^\alpha$ can only have eigenvalues ± 1 . As they mutually anti-commute, none of them can be proportional to the identity, hence each one has the spectrum $\{-1, +1\}$. Let $|\uparrow\rangle$ be an eigenvector of $\tilde{\sigma}^z$ with eigenvalue 1, and construct $|\downarrow\rangle := \tilde{\sigma}^x |\uparrow\rangle$. We find

$$\tilde{\sigma}^z |\downarrow\rangle = \tilde{\sigma}^z \tilde{\sigma}^x |\uparrow\rangle = -\tilde{\sigma}^x \tilde{\sigma}^z |\uparrow\rangle = -\tilde{\sigma}^x |\uparrow\rangle = -|\downarrow\rangle \quad (2.3)$$

and

$$\tilde{\sigma}^x |\downarrow\rangle = (\tilde{\sigma}^x)^2 |\uparrow\rangle = |\uparrow\rangle, \quad (2.4)$$

hence in the basis $\{|\uparrow\rangle, |\downarrow\rangle\}$, $\tilde{\sigma}^\alpha$ can be written as matrices

$$\tilde{\sigma}_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \tilde{\sigma}_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \text{and} \quad \tilde{\sigma}_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad (2.5)$$

where the matrix representation for $\tilde{\sigma}^y$ is fixed by the condition $\tilde{\sigma}^z \tilde{\sigma}^x = i \tilde{\sigma}^y$. Thus, the $\tilde{\sigma}^\alpha$ agree with the usual Pauli matrices in a suitable basis, which means they have to be related to each other by a unitary transformation. $\square$

Bibliography

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